

Title Page



EXCEL TRANSFER TOOL

TECHNICAL DOCUMENTATION & USER GUIDE



VERSION

1.0 April 2026

Parallel Transfer · Ignore Rows · Cell Selector · Fill All Months

Qt 6

C++17

OpenXML

Multi-Thread

Finance Automation Team • Internal Use Only

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Excel Transfer Tool is a high-performance Qt/C++ desktop application that automates monthly cost-control reporting workflows. It loads Excel workbooks (XLSM/XLSX) for multiple periods, applies configurable transfer mappings, consolidates data across months, and produces clean output files—all without manual copy-paste work.

Key Capabilities — v1.0

- Load multiple cost-control files with parallel processing (2-3x faster)
- Apply row-level transfer mappings per month/sheet/column
- Rolling Transfer (RT) mode for sequential month-to-month workflows
- Parallel Transfer — run Individual or Rolling modes from a unified backend
- Fill All Months — backfill Jan through target month in one operation
- Ignore Rows — skip specific rows per mapping card
- Create Month File — generate new month files from existing templates
- Visual Cell Selector — pick source/destination cells from a grid view
- Log Panel toggle — show/hide the status log on demand
- Quarter quick-select buttons (Q1–Q4) for rapid period selection
- Automatic formula preservation and shared formula handling

Feature	Description
Parallel Transfer Backend	New <code>ParallelTransferController</code> with <code>IndividualTransferWorker</code> and <code>RollingTransferWorker</code> . Each worker owns its own <code>ExcelHandler</code> and <code>TransferService</code> — fully thread-safe.
Fill All Months	Writes Jan through target month into a single cost-control file. Destination always points to the <code>Month/</code> subfolder.
Ignore Rows Dialog	Per-mapping-card dialog showing all <code>rowMap</code> entries. Check rows to skip during transfer. Filter by row number or value.
Create Month File	Button on each <code>YearCard</code> . Activates creation mode: select months, click Load Periods, app finds last existing file and copies it for missing months.
Visual Cell Selector	“Select Source/Dest Cells” opens a scrollable grid of the loaded Excel sheet. Click any cell to capture its reference.
Log Panel Toggle	“Log” button in header shows/hides the status log dock. Hidden by default on startup.
Copy Full Sheet hidden	For <code>REFI/Budget/prev-year</code> mappings (same-file sources), “Copy full sheet” is automatically hidden.
Run/Remove buttons removed	Mapping row cards no longer show Run and Remove buttons.
Toast positioning fixed	Blue notification toasts now appear at the bottom-center of the application window, not the screen.

System Requirements

Component	Requirement
Operating System	Windows 10 or Windows 11 (64-bit)
Microsoft Excel	Excel 2016, 2019, 2021, or Microsoft 365
Runtime	Qt 6.x (bundled with executable)
Disk Space	Minimum 200 MB free
RAM	8 GB minimum, 16 GB recommended for 12+ months

Setup Steps

1. Extract the application folder to your preferred location.
2. Ensure `repair.ps1` is in the same folder as `ExcelTransfer.exe`.
3. Launch `ExcelTransfer.exe`.
4. Verify the default folder paths point to your network share (L: drive).
5. Ensure cost-control files follow the expected naming convention.
6. Verify JSON mapping files exist in the JSON/ subfolder.

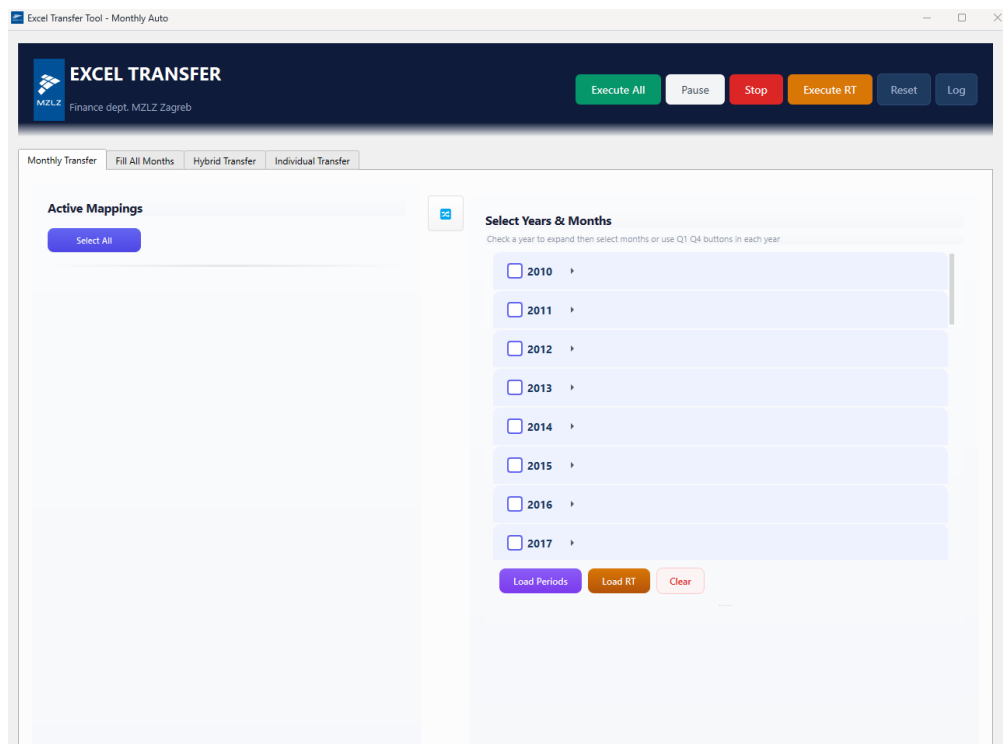


Figure 1: Excel Transfer Tool — Main window showing the Monthly Transfer tab with collapsed year cards.

How to Use the Main Window

Step 1: Understanding the Layout

- 1. The **left sidebar** shows “Active Mappings” — this is where mapping cards appear after loading files.
- 2. The **center panel** displays year cards (2025, 2026, etc.) with month checkboxes.
- 3. The **header toolbar** contains Execute All, Pause, Stop, Reset, and Log buttons.
- 4. The **bottom status bar** shows the current operation status and file counts.

Step 2: First-Time Setup

- 1. Verify the application is pointing to the correct network drive (default: L: drive).
- 2. Ensure all JSON mapping files exist in the JSON/ subfolder next to the executable.
- 3. Check that cost-control files follow the naming convention: **Cost Control ZAG MM_YYYY_working.xlsm**.

Before You Start

Always close all Excel files before loading them in the application. Open files will cause “ZIP read error” or “file is locked” messages and prevent transfers from running.

Header Bar

The header bar contains global controls:

Button	Action
Reset	Clears all loaded files, mapping cards, and period rows
Create Month File	Opens creation mode on the year cards (see Section 11)
Log	Toggles the status log panel (hidden by default)



Figure 2: The **application header bar** with the global toolbar buttons highlighted. From left to right: **Execute All** (green) — runs all checked mapping cards in sequence; **Pause** (white) — suspends the current transfer after the active cell write completes; **Stop** (red) — cancels execution immediately; **Execute RT** (amber) — starts a Rolling Transfer chain using the *Load RT* period; **Reset** (dark) — clears all loaded files, mapping cards, and period selections; **Log** (dark) — toggles the status log panel. The four tabs below give access to *Monthly Transfer*, *Fill All Months*, *Hybrid Transfer*, and *Individual Transfer*.

Understanding the Toolbar Buttons

Execute All (Green Button)

- **When to use:** After loading files and checking the mapping cards you want to transfer.
- **What it does:** Processes all checked mapping cards in sequence, writing data from source files to destination files.
- **Watch out:** Maximum 5 months can be selected at once. More than 5 will show a warning.

Pause (White Button)

- **When to use:** During a transfer, if you need to temporarily halt execution.
- **What it does:** Pauses after the current cell write completes. Click again to resume.
- **Watch out:** The current mapping will finish before pausing — it doesn't stop mid-cell.

Stop (Red Button)

- **When to use:** To cancel a transfer operation immediately.
- **What it does:** Terminates execution. The current sheet finishes saving first to prevent corruption.
- **Watch out:** Partial data may be written. Check the log to see which mappings completed.

Reset (Dark Button)

- **When to use:** To clear all loaded files and start fresh.
- **What it does:** Unloads all workbooks, clears mapping cards, and resets period selections.
- **Watch out:** This does NOT delete files — it only clears the application's memory.

Log (Dark Button)

- **When to use:** To view detailed transfer logs and debug information.
- **What it does:** Shows/hides the log panel at the bottom of the window.
- **Watch out:** The log panel is hidden by default. Click to show it when troubleshooting.

Year Cards

Each year card shows month checkboxes and quick-select controls:

- **Year checkbox** — expands/collapses the card
- **Q1–Q4 buttons** — select/deselect entire quarters
- **All / None** — select or deselect all 12 months
- **+ Create Files** — activates file creation mode (amber button, appears when card is expanded)

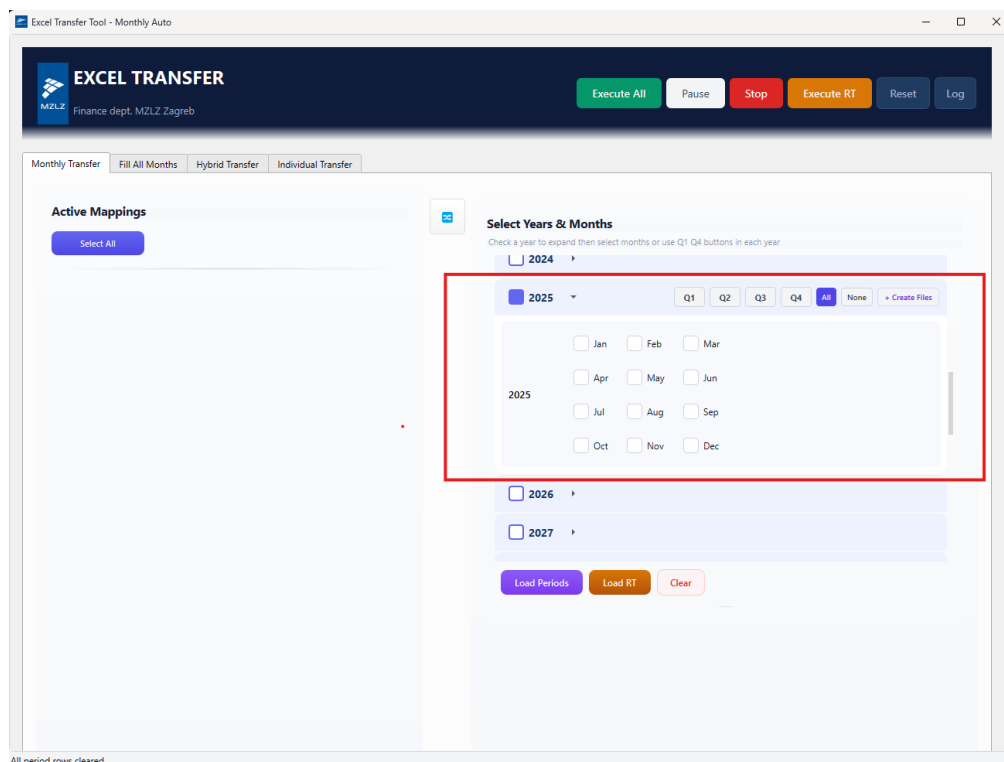


Figure 3: Year card expanded showing month checkboxes. July, August and September are selected (blue). Q1–Q4, All, and None quick-select buttons are visible on the right.

How to Select Months

Step 1: Expand a Year Card

1. Click the **year checkbox** (e.g., “2025”) to expand the card.
2. The card opens to show 12 month checkboxes (January through December).
3. Quick-select buttons appear on the right: Q1, Q2, Q3, Q4, All, None.

Step 2: Select Individual Months

1. Click **individual month checkboxes** to select specific months (e.g., July, August, September).
2. Selected months turn **blue** with a checkmark.
3. You can select months from multiple years by expanding multiple year cards.

Step 3: Use Quick-Select Buttons

- **Q1:** Selects January, February, March
- **Q2:** Selects April, May, June
- **Q3:** Selects July, August, September
- **Q4:** Selects October, November, December
- **All:** Selects all 12 months
- **None:** Deselects all months in this year

Step 4: Load the Selected Periods

1. After selecting months, click **LOAD Selected Periods** at the bottom.
2. Wait for the green “Loaded” toast notification.

Selection Limit

Maximum 5 months can be selected for Execute All. Selecting more will show a warning: “You have selected X months. Maximum is 5 at a time.” Process months in batches to avoid memory overflow.

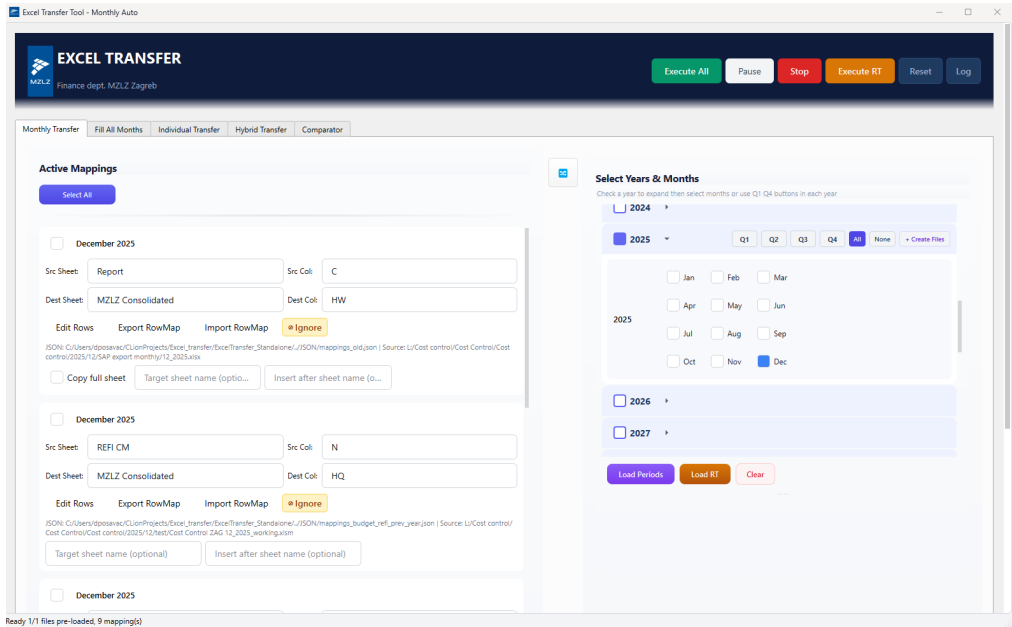


Figure 4: **Monthly Transfer** tab with multiple months selected across different years. The year cards for 2025 and 2026 are both expanded, showing individual month checkboxes. The **Load Selected Periods** button at the bottom becomes active once at least one month is checked. The status bar indicates how many periods are currently selected.

Working with Multiple Years

When to Select Multiple Years

- When you need to process December 2025 and January 2026 together.
- When backfilling data across a year boundary.
- When comparing or consolidating data from different fiscal years.

How to Select Across Years

1. Expand the first year card (e.g., 2025) and select months (e.g., November, December).
2. Expand the second year card (e.g., 2026) and select months (e.g., January, February).
3. The status bar shows total selected: “4 periods selected”.
4. Click **LOAD Selected Periods** to load all months from both years.

What Happens During Load

1. The application scans for cost-control files in each month’s folder.
2. For each found file, it loads the workbook structure (sheets, columns, rows).
3. Mapping cards appear in the left sidebar — one set per month.
4. A green toast notification confirms: “Loaded 4 period(s)”.

File Naming Across Years

Ensure cost-control files follow the naming convention for both years:

Cost Control ZAG 12_2025_working.xlsm

Cost Control ZAG 01_2026_working.xlsm

The application automatically adjusts paths based on the year and month.

Period Limit

Maximum **5 months** can be selected at a time for Execute All. Selecting more shows a warning and prevents execution to avoid memory overflow.

Log Panel

Click the **Log** button in the header to show or hide the status log dock. The log panel is hidden by default on startup. When visible, it shows all transfer status, warnings, and debug output in real time.

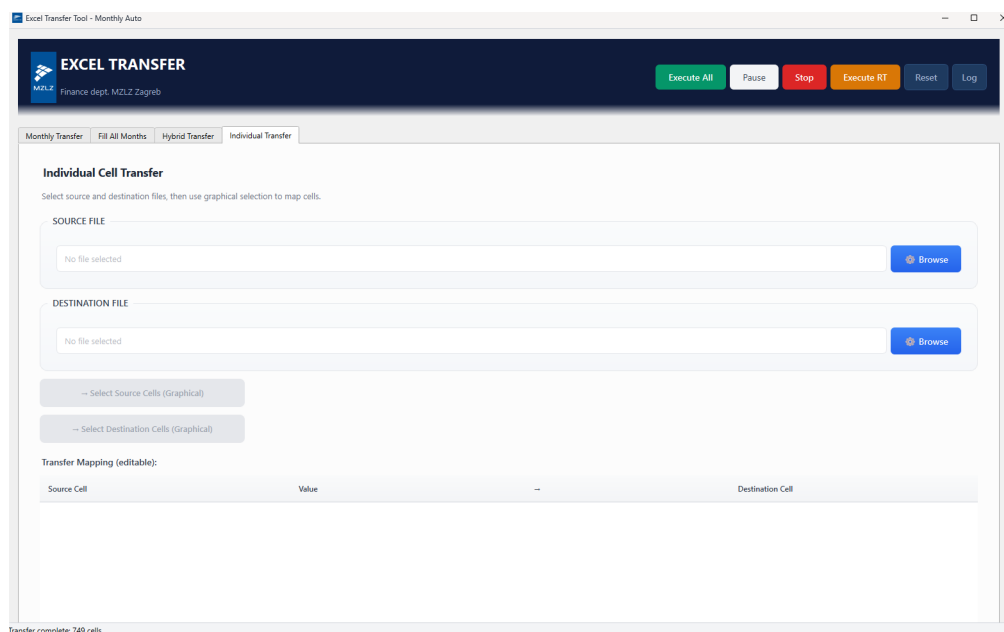


Figure 5: **Individual Transfer** tab in its initial state. Browse for a *Source File* and *Destination File* using the **Browse** buttons. Once files are loaded, the **Select Source Cells (Graphical)** and **Select Destination Cells (Graphical)** buttons become active for visual cell mapping. The *Transfer Mapping* table at the bottom shows all defined source-to-destination cell pairs ready for execution.

Selecting Periods

1. Check the year checkbox (e.g. 2025) to expand the year card.
2. Select months individually or use Q1–Q4 / All / None buttons.
3. Click **LOAD Selected Periods**.
4. Wait for the green “Loaded” toast notification.

Important

Ensure all target files are **closed in Excel** before loading. Open files cause ZIP read errors.

What is a Mapping Card?

Each mapping card represents a transfer rule:

- A **source** workbook, sheet, and column

- A **destination** sheet and column
- A **rowMap** defining source row → destination row pairs

Mapping Card Buttons

Button	Action
Edit Rows	Modify source/destination row numbers
Export	Export rowMap to JSON file
Import	Import rowMap from JSON file
 Ignore	Open Ignore Rows dialog to skip specific rows

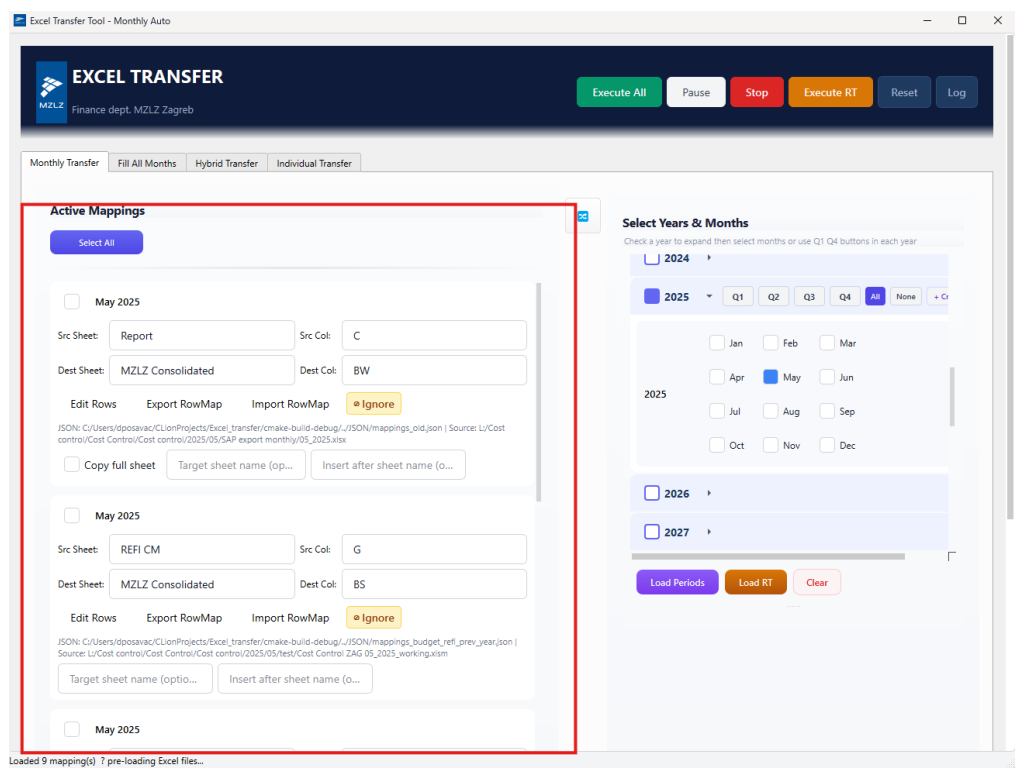


Figure 6: **Active Mappings** sidebar on the Monthly Transfer tab, showing loaded mapping cards for May 2025. Each card displays the source sheet, source column, destination sheet, and destination column. Cards with an amber **Ignore** badge are excluded from the next transfer run. The **Copy full sheet**, **Target sheet name**, and **Insert after sheet name** fields appear below cards that support full-sheet copying. The year/month selector on the right shows May 2025 highlighted.

Understanding Mapping Cards

What is a Mapping Card?

- Each card represents one transfer rule: source → destination.
- **Source:** Which Excel file, sheet, and column to read from.
- **Destination:** Which sheet and column to write to.
- **Row Map:** Defines which source rows map to which destination rows.

Reading a Mapping Card

1. **Top line:** Source file type (e.g., “SAP Report”, “2025 FTE”, “REFI CM”).
2. **Second line:** Source sheet name and column (e.g., “Report C”).
3. **Third line:** Destination sheet and column (e.g., “MZLZ Consolidated BD”).
4. **Checkbox:** Check to include this mapping in the next Execute All run.
5. **Amber badge:** Shows “Ignore” if this mapping is temporarily disabled.

Card Actions

- **Edit Rows:** Modify which source rows map to which destination rows.
- **Export:** Save the current rowMap to a JSON file for backup or sharing.
- **Import:** Load a rowMap from a JSON file to replace the current mapping.
- **Ignore badge:** Click to toggle this mapping in/out of transfers (turns amber when ignored).

Copy Full Sheet Option

- Appears below cards that support full-sheet copying (e.g., external source files).
- **Check “Copy full sheet”** to copy the entire source sheet instead of individual cells.
- **Target sheet name:** Specify the name for the copied sheet in the destination file.
- **Insert after sheet name:** Position the new sheet after an existing sheet.

Hidden for Same-File Sources

The “Copy full sheet” option is **automatically hidden** for REFI, Budget, and prev-year mappings because these read from the same cost-control file. Copying a sheet to itself is not applicable.

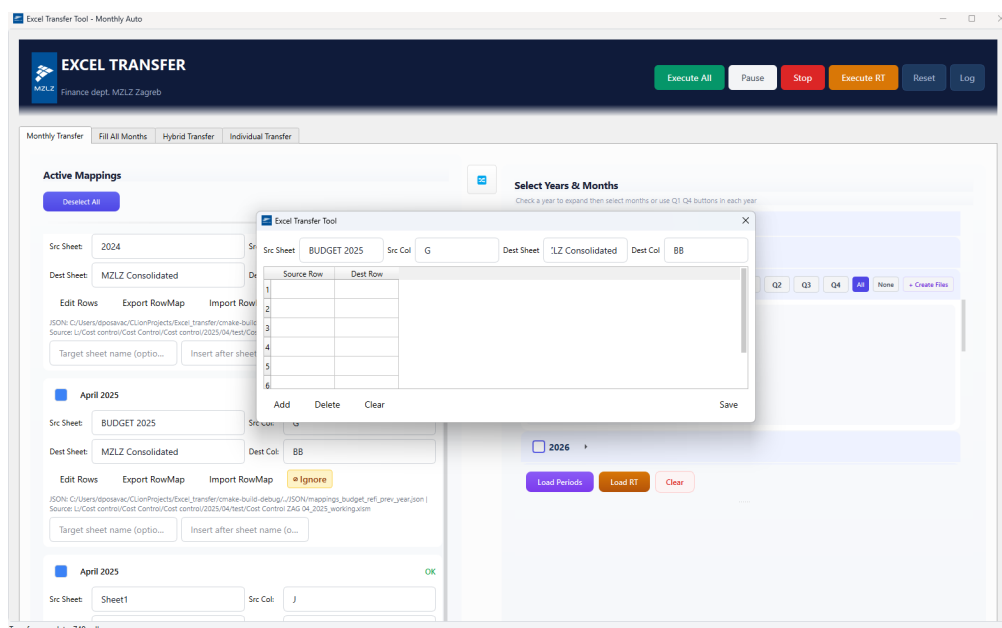


Figure 7: **Mapping card detail view** showing the expanded controls for a single mapping entry. The card displays the source file type (SAP Report), source sheet name, source column (C), destination sheet (MZLZ Consolidated), and destination column (BD). Below the main mapping information are action buttons: **Edit Rows** (modify row mappings), **Export** (save rowMap to JSON), **Import** (load rowMap from JSON), and the amber **Ignore** badge (toggle this mapping in/out of transfers). The checkbox at the top left controls whether this mapping is included in the next Execute All run.

Working with Mapping Card Controls

Checkbox (Top Left)

1. **Checked (blue)**: This mapping will be included in the next Execute All run.
2. **Unchecked (gray)**: This mapping will be skipped during Execute All.
3. **Use case**: Uncheck mappings you don't need to run (e.g., skip SAP if only updating PAX data).

Edit Rows Button

1. Click **Edit Rows** to open the row mapping editor.
2. The dialog shows a table: left column = source rows, right column = destination rows.
3. **Add new mappings**: Click "+ Add Row" and enter source/destination row numbers.
4. **Delete mappings**: Select a row and click "Delete".
5. **Bulk operations**: Use "Import" to load from JSON or "Export" to save.
6. Click **OK** to apply changes.

Export Button

1. Click **Export** to save the current rowMap to a JSON file.
2. Choose a location and filename (e.g., `my_custom_mapping.json`).
3. **Use case**: Backup your custom mappings or share them with colleagues.

Import Button

1. Click **Import** to load a rowMap from a JSON file.
2. Select a previously exported JSON file.

3. The current rowMap is replaced with the imported one.

4. **Watch out**: This overwrites the existing mapping — export first if you want to keep

Import Overwrites Data

Importing a rowMap **replaces** the current mapping completely. Always export your current mapping first if you want to preserve it.

Copy Full Sheet

The “Copy full sheet” checkbox is **automatically hidden** for REFI, Budget, and prev-year mappings. These mappings read from the same cost-control file — copying a sheet to itself is not applicable.

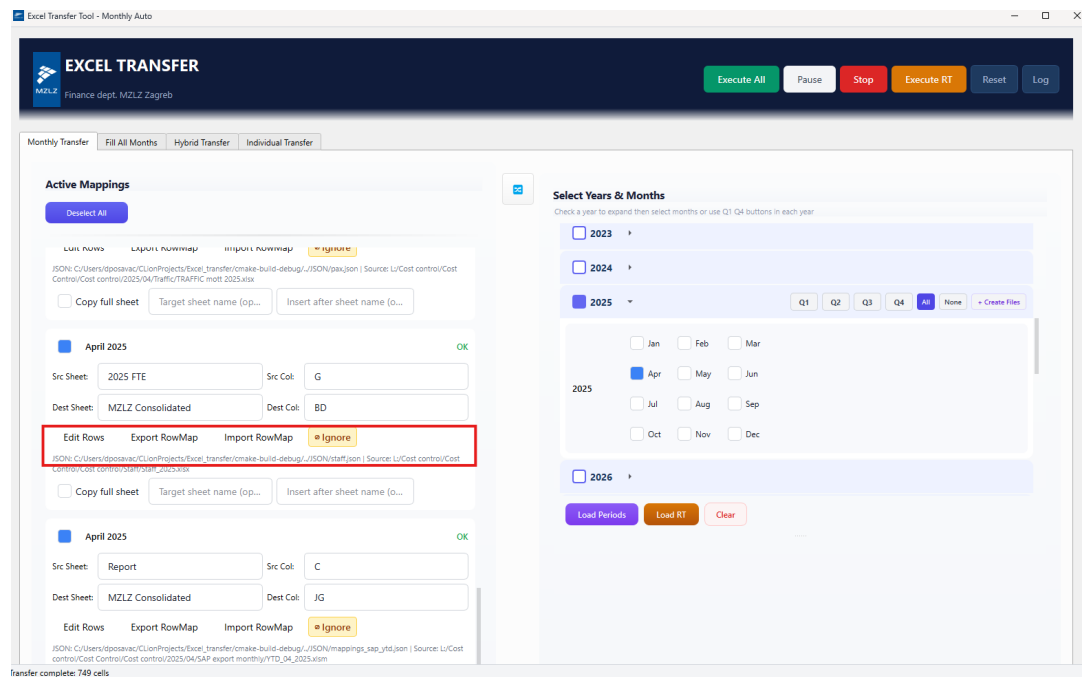


Figure 8: **Active Mappings** sidebar showing a mapping card for April 2025 with the **Ignore** badge highlighted (amber). The card maps the *2025 FTE* source sheet (column G) to *MZLZ Consolidated* (column BD). Clicking the **Ignore** badge on any card toggles it in or out of the next transfer run without removing the mapping. The **Copy full sheet** checkbox and sheet name fields below the card allow full-sheet copy mode for supported mapping types.

Ignore Rows Dialog

Click  **Ignore** on any mapping card to open the Ignore Rows dialog:

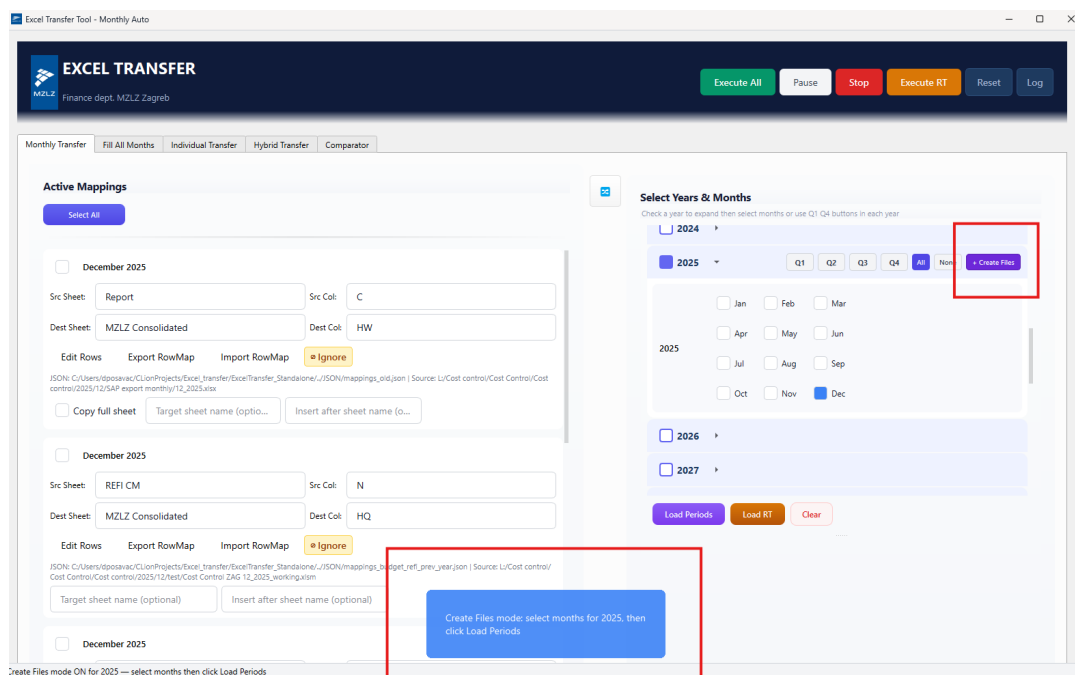


Figure 9: **Edit Rows dialog** for modifying source and destination row mappings. The dialog shows a two-column table where each row defines a mapping pair: the left column lists source row numbers (from the source Excel file), and the right column lists corresponding destination row numbers (in the destination file). Users can add new row pairs using the **+ Add Row** button, delete existing pairs with the **Delete** button, or bulk-import/export mappings using the **Import** and **Export** buttons. Changes are applied immediately when clicking **OK**.

1. The dialog shows a table of all rowMap entries: Dest Row, Dest Column, Src Row(s), Current Value.
2. **Check the “Skip” checkbox** next to any row you want to exclude from transfer.
3. Use the **Filter** box to search by row number or cell value.
4. Click **Ignore All** / **Ignore None** for bulk selection.
5. Click **Apply** to save. The status shows how many rows will be skipped.

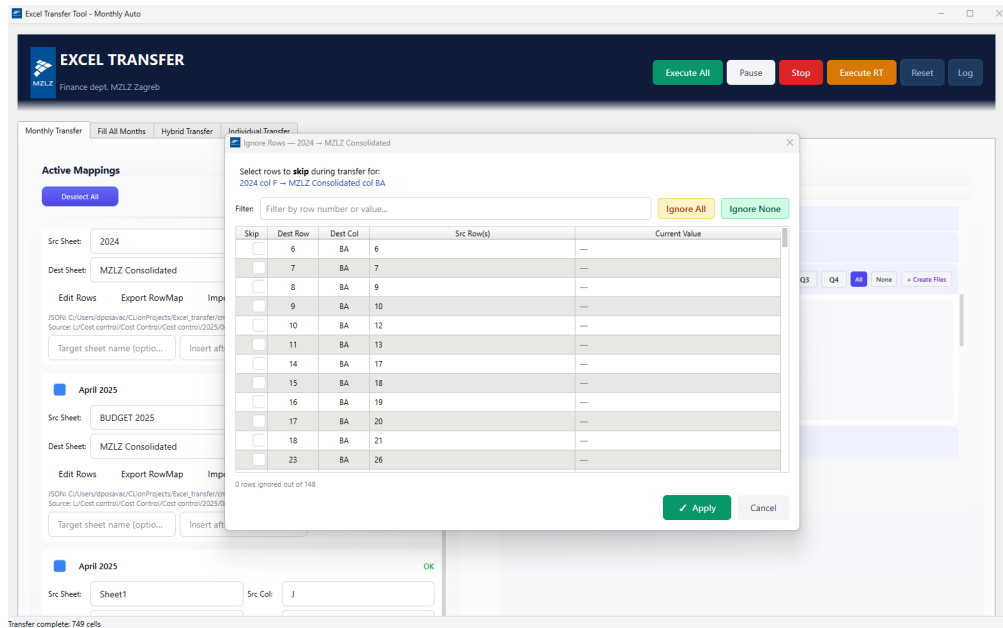


Figure 10: Ignore Rows dialog for REFI CM → MZLZ Consolidated. Each rowMap entry shows Dest Row, Dest Column, Src Row(s), and Current Value. Check **Skip** to exclude a row from the next transfer. The filter box narrows the list. Status line shows “0 rows ignored out of 136”.

Using the Ignore Rows Dialog

Step 1: Open the Dialog

1. On any mapping card, click the **Ignore** button.
2. The Ignore Rows dialog opens, showing all rowMap entries for this mapping.
3. Each row displays: Dest Row, Dest Column, Src Row(s), and Current Value.

Step 2: Review the Row List

- **Dest Row:** The destination row number in the target Excel file.
- **Dest Column:** The destination column (e.g., BD, CP, DJ).
- **Src Row(s):** The source row(s) that map to this destination (can be multiple).
- **Current Value:** The current value in the destination cell (if file is loaded).
- **Skip checkbox:** Check to exclude this row from the next transfer.

Step 3: Select Rows to Ignore

1. **Manual selection:** Check the “Skip” checkbox next to any row you want to exclude.
2. **Filter:** Type in the filter box to search by row number or cell value (e.g., “115” or “1000”).
3. **Bulk selection:** Click **Ignore All** to check all rows, or **Ignore None** to uncheck all.
4. The status line updates: “X rows ignored out of Y”.

Step 4: Apply Changes

1. Click **Apply** to save your selections.
2. The dialog closes and returns to the main window.
3. Ignored rows will be skipped during the next Execute All run.
4. Click **Cancel** to discard changes and close the dialog.

When to Use Ignore Rows

- **Testing:** Ignore most rows to test a transfer with just a few rows.
- **Selective updates:** Skip rows that shouldn’t be overwritten (e.g., manual adjustments).

Session Only

Ignored rows are stored in memory for the current session only. They are **not saved** to JSON files and will reset when you restart the application. If you need permanent exclusions, modify the JSON mapping files directly.

Filter Tips

The filter box searches both row numbers and cell values:

- Type “115” to find all rows with 115 in the row number.
- Type “1000” to find all rows with values containing 1000.
- Type “BD” to find all rows in column BD.

Complete Workflow: From Loading to Execution

Step 1: Load Files

1. Expand year cards and select months (maximum 5 at a time).
2. Click **LOAD Selected Periods**.
3. Wait for the green toast: “Loaded X period(s)”.
4. Verify mapping cards appear in the left sidebar.

Step 2: Review Mapping Cards

1. Check that all expected mapping cards are present (typically 9 per month: SAP, REFI, Budget, Traffic, PAX, Staff, SAP YTD, PAX Transfer, Prev Year).
2. **Uncheck** any mappings you don’t want to run.
3. Click  **Ignore** on cards where you want to skip specific rows.
4. Verify source/destination columns are correct.

Step 3: Execute the Transfer

1. Ensure all destination files are **closed in Excel**.
2. Click the green **Execute All** button in the header toolbar.
3. A progress dialog appears showing the current operation.
4. Monitor the progress bar and status messages.

Step 4: Monitor Progress

1. The progress dialog shows: “Processing [Source] [Destination]”.
2. The progress bar updates as each mapping completes.
3. Click **Log** in the header to view detailed execution logs.
4. Watch for any red error messages in the log panel.

Step 5: Review Results

1. When complete, a summary dialog shows: total mappings, successful transfers, failed transfers, and cells transferred.
2. A green toast notification appears: “Transfer complete — X cells transferred”.
3. Check the log panel for any warnings or errors.
4. Open the destination Excel files to verify the data was written correctly.

Common Errors**“ZIP read error” or “not a zip file”**

- **Cause:** File is open in Excel, corrupt, or not a valid XLSX/XLSM file.
- **Fix:** Close the file in Excel. If still failing, replace with a fresh copy from backup.

“0 cells transferred”

- **Cause:** Source file missing, sheet not found, or all rows are ignored.
- **Fix:** Check the log for specific error messages. Verify file paths and sheet names.

“File is locked”

- **Cause:** Destination file is open in Excel or locked by another process.
- **Fix:** Close the file in Excel. Check Task Manager for orphaned Excel processes.

Pause and Stop

- **Pause** — temporarily halts execution (can be resumed)
 - **Stop** — terminates execution (current sheet finishes first)
- 

What is Rolling Transfer?

Rolling Transfer processes months **sequentially**, where each month’s output becomes the input for the next. Use RT when building month-to-month cumulative data chains.

How RT Works

1. Select multiple months in chronological order.
2. Click **Load RT** to validate source files exist.
3. Click **Execute RT**:
 - Copies previous month file to new month’s **Month/** folder
 - Applies all mappings (traffic_mott, pax_transfer, pax, staff, SAP, REFI, sap_ytd)
 - Saves and reloads before proceeding to next month

RT Error Handling

RT **aborts** on the first error. Because each month depends on the previous output, continuing after an error would produce incorrect cumulative data.

What is Fill All Months?

Fill All Months writes data for **all months January through a target month** into a single cost-control file. Use this to backfill an entire year's worth of data into one destination file.

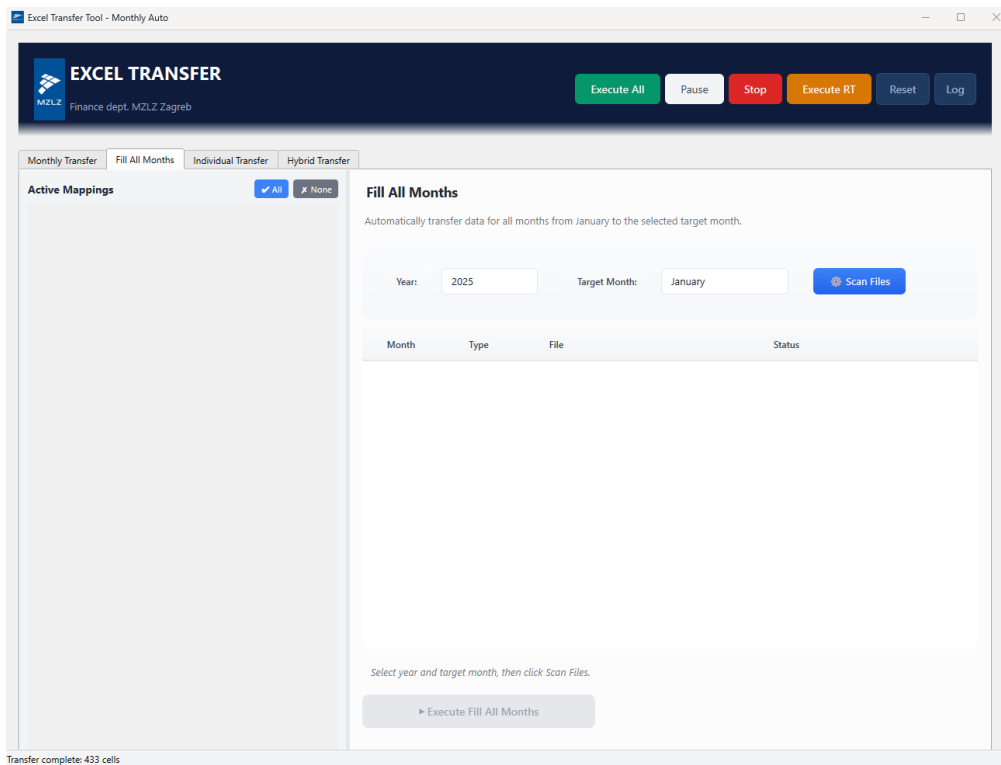


Figure 11: **Fill All Months** tab in its initial state. The left sidebar shows the *Active Mappings* panel with **All** and **None** toggle buttons for selecting which mapping types to include. On the right, set the *Year* and *Target Month*, then click **Scan Files** to discover all source files from January through the target month. The *Execute Fill All Months* button becomes active after a successful scan.

How it Works

1. Navigate to the **Fill All Months** tab.
2. Select the **year** and **target month** (e.g. November 2025).

- 3. Click **Scan Files** — the app scans for all source files (SAP, PAX, Staff, etc.) for each month Jan through November.
- 4. Review the scan table: green checkmarks show found files, red X marks show missing files.
- 5. Click **Execute Fill All Months** to run the transfer.

Destination Path

The destination file is always in the **Month/** subfolder:
L:/Cost control/.../2025/11/Cost Control ZAG 11_2025_working.xlsm

Corrupt Files

If the destination file shows “not a zip file” error, it may be corrupt. Replace it with a fresh copy from the backup folder before running Fill All Months.

Overview

The Parallel Transfer system is a new backend that provides two independent transfer modes, each running on its own thread with its own **ExcelHandler** and **TransferService** instances:

Mode	Behaviour
Individual	Each month runs independently. On error: logs and continues to next month. No save+reload between months.
Rolling	Months run sequentially in calendar order. On error: aborts remaining months (data integrity). Save+reload after each month so next month reads fresh output.

Key Design Principles

- **Thread safety** — each worker owns its own handler/service, never shared
- **Decoupled** — deleting the parallel transfer files does not affect existing functionality
- **Validation** — Rolling requires ≥ 2 contiguous months before thread is created
- **Cancellable** — both workers check `m_cancelled` flag between entries

Rolling Validation Rules

The Rolling worker validates before starting:

- At least **2 months** must be selected
 - Months must be **contiguous** (no gaps, e.g. Jan + Mar without Feb is rejected)
 - Validation runs on the **UI thread** (instant feedback) and again on the **worker thread** (safety)
-

How it Works

The Create Month File feature copies an existing cost-control xlsx as a template for new months:

1. Expand a year card (e.g. 2025).
2. Click the amber + **Create Files** button.
3. Select the months you want to create (e.g. May, June, July).
4. Click **Load Periods** (now in creation mode).
5. The app automatically finds the **last existing** cost-control file for that year (scans December → January).
6. A confirmation dialog shows the source file and destination paths.
7. Click **Yes** to create the files.

Source File Selection

The app always copies from the **most recent existing** month. If only February exists, it uses February for March, April, May, etc. Files are created in the **Month/** subfolder of each month's directory.

Result

A toast notification confirms: "Created 3 file(s) from February 2025." Already-existing files are skipped (not overwritten).

When configuring an **Individual Transfer**, click **Select Source Cells** or **Select Destination Cells** to open a visual cell picker:

- **Sheet dropdown** at the top — switch between sheets
- **Scrollable grid** showing cell data (rows 1–50, columns A–Z)
- **Click any cell** — automatically fills the Column and Row fields
- **“Selected:” label** updates in real time (e.g. `Sheet1!N17`)
- **Manual override** — type column letter and row number directly
- Click **Apply** to confirm selection

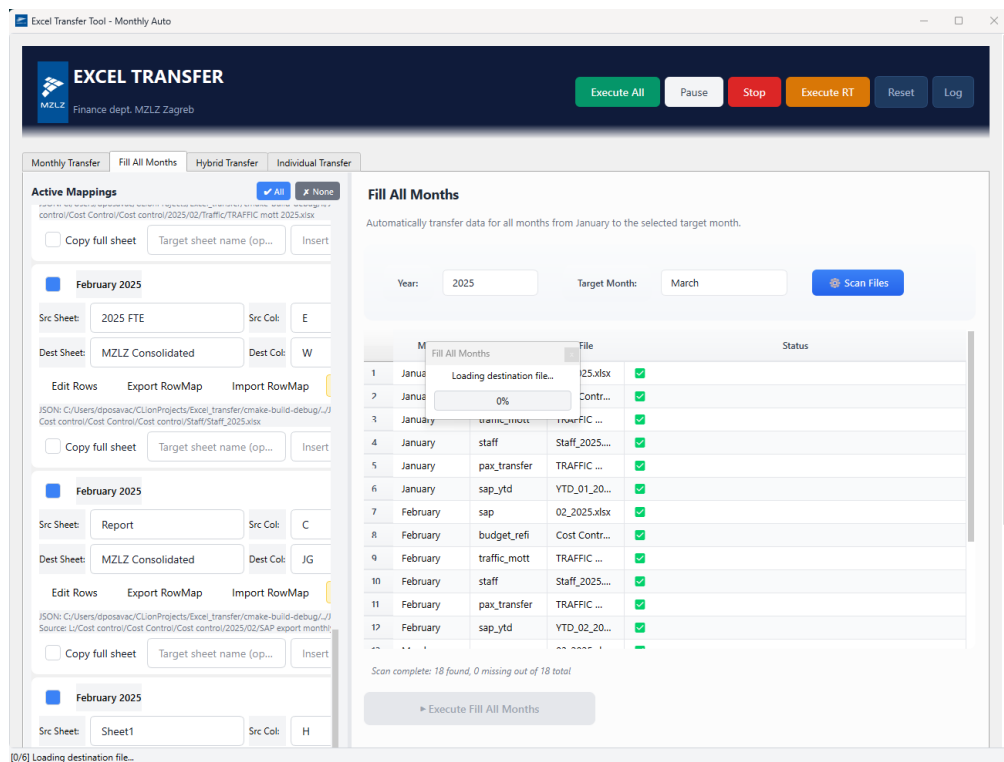


Figure 12: **Fill All Months** tab while a transfer is in progress for year 2025, target month March. The left sidebar lists *Active Mapping* cards for each discovered month (January, February shown). The scan results table on the right lists all found source files per month and mapping type (sap, budget_refi, traffic_mott, staff, pax_transfer, sap_ytd), each confirmed with a green checkmark. A progress dialog shows the current file loading stage at 0%.

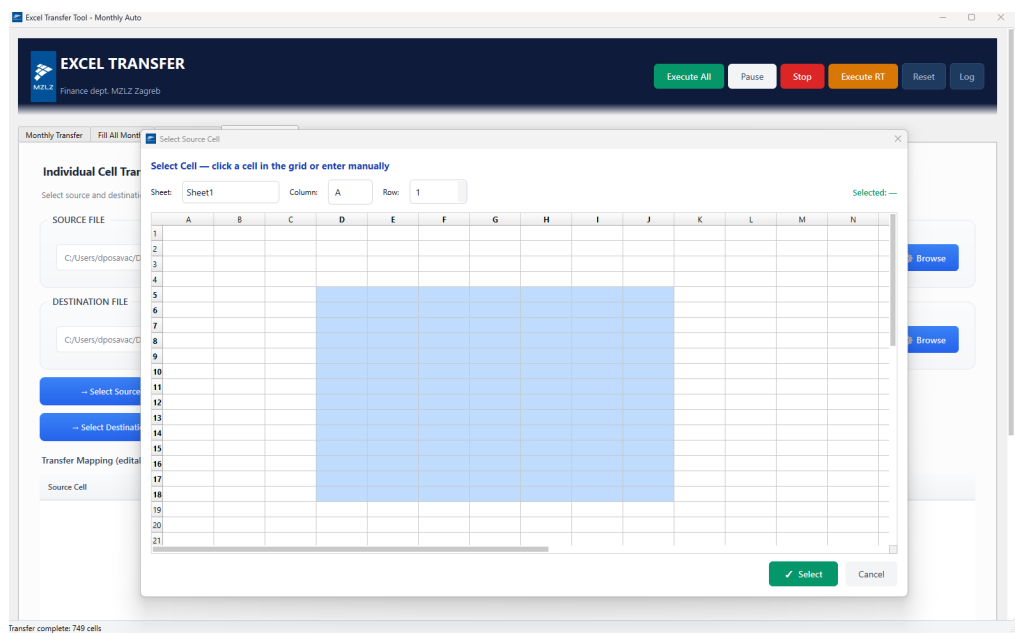


Figure 13: Visual Cell Selector dialog — “Select Source Cell”. The grid shows the loaded workbook data (Sheet1, columns A–N, rows 1–21). Click any cell to capture its reference. The Sheet, Column, and Row fields update automatically. The highlighted area shows existing data in the source file.

File Must Be Loaded

The cell selector reads data from the already-loaded workbook. If the source or destination file has not been loaded, the grid will be empty. Load the file first using Load Periods.

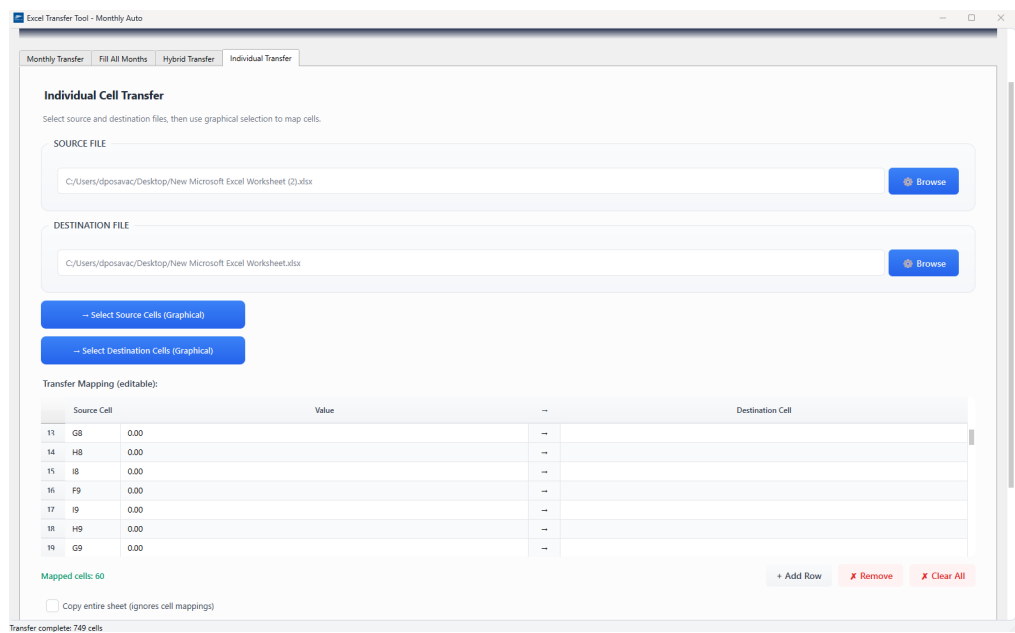


Figure 14: Individual Transfer tab with 105 cells mapped. The Transfer Mapping table lists source cells (D4, C4, I4, H4, F4, E4, G4) with their current values and destination cells. The **Execute Transfer** button becomes active once mappings are configured.

Traffic Mott YTD Flow

Step	Source	Destination
1	External TRAFFIC mott YYYY.xlsx Sheet1	Cost control TRAF- FIC mott YYYY sheet (cols D–O)
2	External TRAFFIC mott YYYY.xlsx Sheet1	Cost control TRAFFIC mott Q (Q9/Q10 month number)
3	External TRAFFIC mott YYYY.xlsx (sum G..currentCol)	MZLZ Consoli- dated JG rows 5,7,8,9,10,11
4	Staff {year} FTE sheet row 33	MZLZ Consoli- dated JG row 16

YTD Calculation

PAX source columns G–R contain **single-month values** (not cumulative). The JG column YTD is computed in C++ by summing columns G through the current month column directly from the external TRAFFIC mott file.

Transfer Execution Order

Mappings execute in this fixed priority order:

Priority	Type	What it does
1	traffic_mott	Writes monthly columns D–O in TRAFFIC mott sheet
2	pax_transfer	Writes PAX values to TRAFFIC mott (same sheet)
3	pax	Writes PAX monthly values to MZLZ (col GX etc.)
4	staff	Writes FTE data from Staff file
5	sap	Writes SAP monthly data to MZLZ
6	budget_refi	Writes REFI, Budget, prev-year columns
7	sap_ytd	LAST: reads Q from TRAFFIC mott, writes JG + FTE

“not a zip file” Error

Symptom

QZip: not a zip file! in log. All 9 mappings fail with 0 cells.

Cause: The cost-control xlsx file is corrupt or was created from a corrupted template.

Fix: Replace the file with a fresh copy from the Backup folder or copy from another month’s working file.

0 Cells Transferred

- Verify destination file is in the **Month/** subfolder
- Check source files (SAP, PAX, Staff) are present and correctly named
- Ensure no rows are accidentally marked as “Ignore” in the Ignore Rows dialog
- Review log panel for “sheet not found” or “file missing” warnings

Stack Overflow / Crash on Execute All

Symptom

Application crashes with exit code 0xC0000602 when clicking Execute All.

Cause: Too many months selected simultaneously creates too many Qt widgets.

Fix: Select **maximum 5 months** at a time. The app will show a warning if more are selected.

Fill All Months Writes to Wrong Folder

Ensure the destination path shown in the scan table ends with **/Month/**. If it points to the root month folder, the file naming in that folder may be non-standard.

Parallel Transfer Not Visible in UI

The Parallel Transfer backend is implemented but the UI tab is under development. The backend can be connected by wiring `ParallelTransferController` signals in `mainwindow.cpp`.

File Naming Convention

Cost Control ZAG {MM}_{YYYY}_working.xlsm

Location: L:/Cost control/Cost Control/Cost control/{YYYY}/{MM}/month/

File Type	Path Pattern
Cost Control	.../{YYYY}/{MM}/Cost Control ZAG {MM}_{YYYY}_working.xlsm
SAP Monthly	.../{YYYY}/{MM}/SAP export monthly/{MM}_{YYYY}.xlsx
SAP YTD	.../{YYYY}/{MM}/SAP export monthly/YTD_{MM}_{YYYY}.xlsm
Traffic (PAX)	.../{YYYY}/{MM}/Traffic/TRAFFIC mott {YYYY}.xlsx
Staff	.../Staff/Staff_{YYYY}.xlsx

JSON Mapping Files

File	Type	Purpose
mappings_old.json	sap	SAP monthly → MZLZ Consolidated
mappings_budget_refi_prev_year.json	budget_refi	REFI/Budget/PrevYear → MZLZ
Traffic_mott.json	traffic_mott	PAX Sheet1 → TRAF-FIC mott sheet
mappings_pax_transfer.json	pax_transfer	PAX Sheet1 → TRAF-FIC mott (extra rows)
mappings_sap_ytd.json	sap_ytd	TRAFFIC mott Q → MZLZ JG
pax.json	pax	PAX Sheet1 → MZLZ monthly PAX cols
staff.json	staff	Staff FTE → MZLZ

Technical Architecture

Component	Technology
UI Framework	Qt 6.x (Widgets)
Language	C++17
Excel Parsing	OpenXML (ZIP + XML), no COM dependency
Compression	miniz (embedded)
Threading	QThread, QThreadPool
JSON Parsing	Qt JSON (QJsonDocument)
Parallel Transfer	ParallelTransferController + IndividualTransferWorker + Rolling-TransferWorker

Performance Characteristics

Operation	Performance
Transfer 1 month (all 9 mappings)	10–60 seconds (depends on file size)
Save 50MB XLSM file	8–20 seconds
Fill All Months (12 months)	3–8 minutes
RT chain (6 months)	2–4 minutes
Maximum safe months (Execute All)	5 months

Hybrid Transfer Tab

The **Hybrid Transfer** tab lets you assign individual months to either *Execute All* (cell-by-cell mapping) or *Execute RT* (rolling transfer), then run both modes automatically in sequence. This is useful when some months need a standard transfer while others require a rolling transfer chain.

How to use Hybrid Transfer:

1. Select the **Year** and set the **Transfer Type** (*Execute All* or *Execute RT*) from the dropdown.
2. Check the months you want to assign using the month checkboxes or the *Q1–Q4 / All / None* quick-select buttons.
3. Click + **Add Selected Months** to add them to the *Assigned Periods* table below.
4. Repeat steps 1–3 to add more months with a different transfer type.
5. Click **Execute All** in the header toolbar to run all assigned periods in sequence.

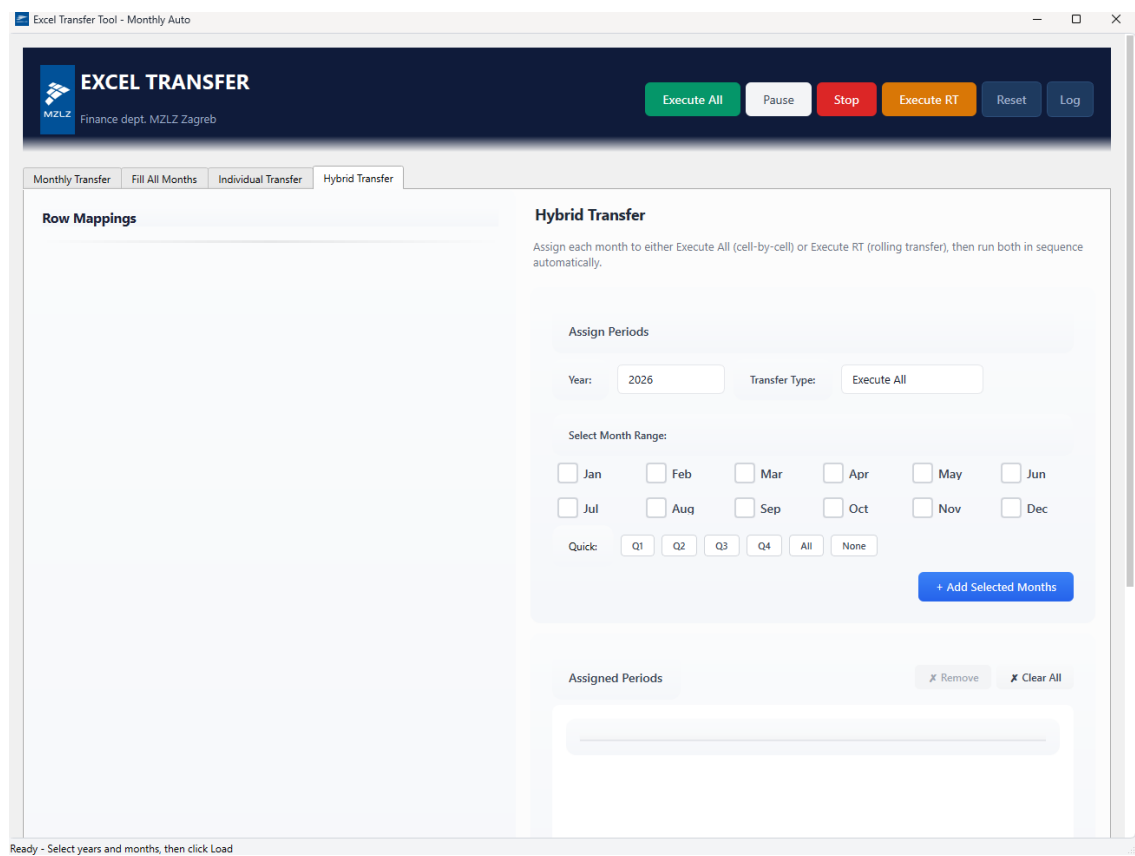


Figure 15: The **Hybrid Transfer** tab in its initial state. The left panel shows *Row Mappings* (mapping cards appear here once months are loaded). On the right, the *Assign Periods* section allows selecting a **Year** (2026 shown) and **Transfer Type** (*Execute All* or *Execute RT*). Individual months are checked via the month grid or the *Q1–Q4 / All / None* quick-select buttons. Click **+ Add Selected Months** to register the selection in the *Assigned Periods* table. The status bar reads “Ready – Select years and months, then click Load”.

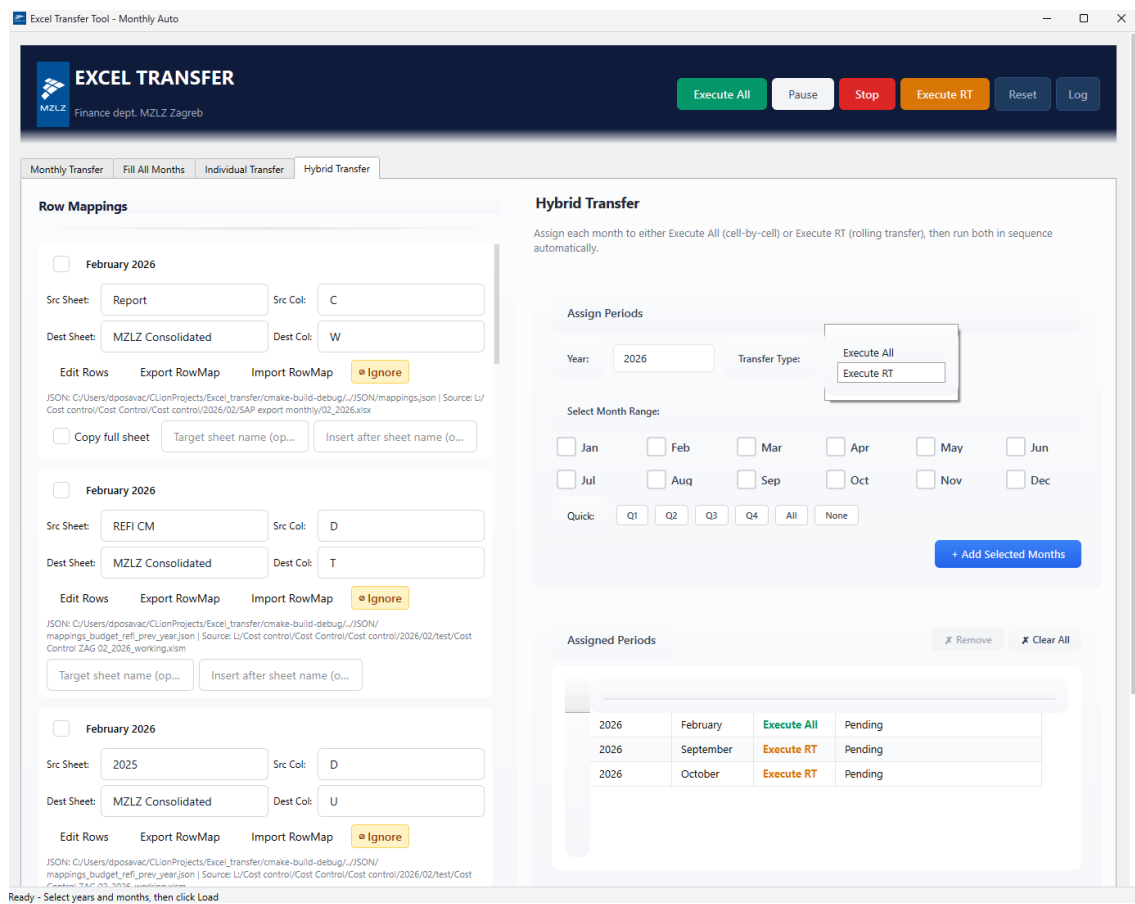


Figure 16: The **Hybrid Transfer** tab with periods assigned and mapping cards loaded for February 2026. The left *Row Mappings* sidebar shows all mapping cards for the loaded month (SAP Report, REFI CM, Budget 2025 — each with source/destination sheet and column, and an amber **Ignore** badge where applicable). The *Transfer Type* dropdown on the right shows both options — *Execute All* and *Execute RT* — open for selection. The *Assigned Periods* table at the bottom lists three entries: February 2026 (*Execute All*, Pending), September 2026 (*Execute RT*, Pending), and October 2026 (*Execute RT*, Pending). Click **Execute All** in the header toolbar to process all assigned periods.

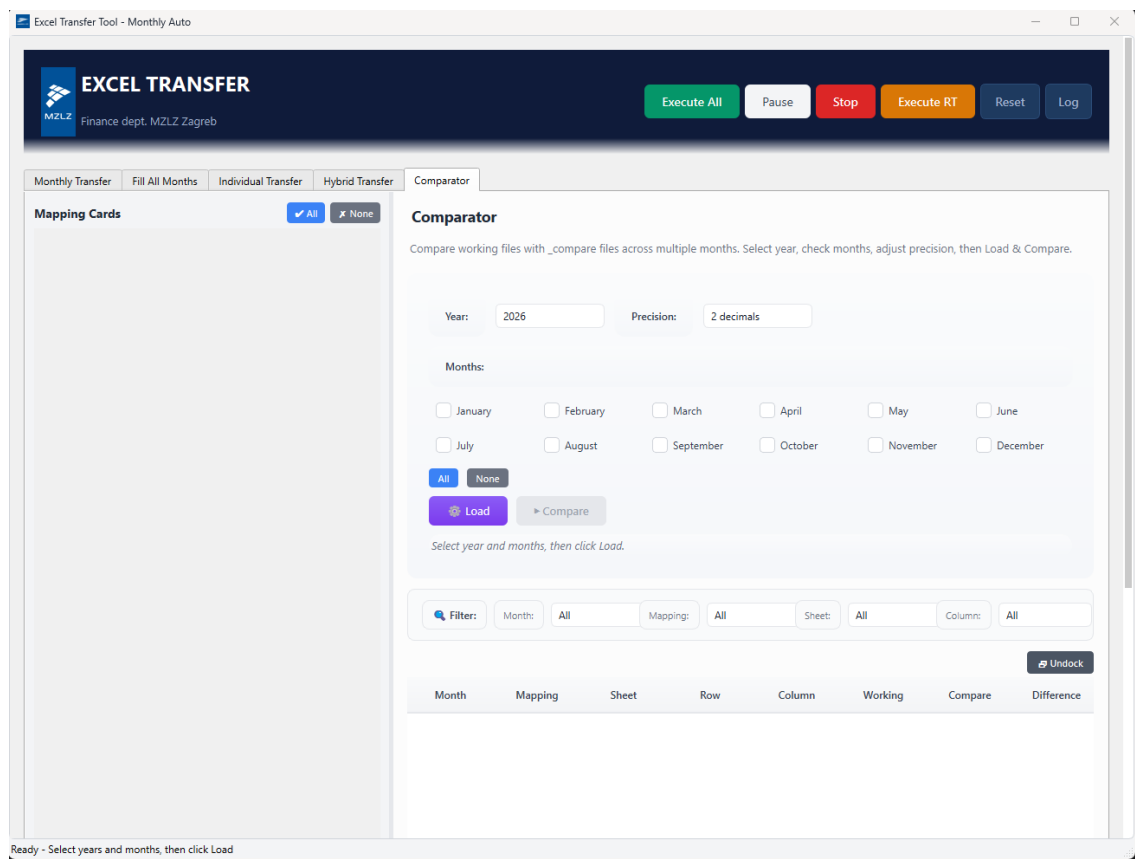


Figure 17: **Transfer progress dialog** showing real-time status during an Execute All operation. The dialog displays the current operation (“Transferring data...”), a progress bar indicating completion percentage, and the current mapping being processed (e.g., “Processing SAP Report MZLZ Consolidated”). The **Cancel** button allows users to abort the transfer mid-execution. Status messages update continuously as each mapping card completes.

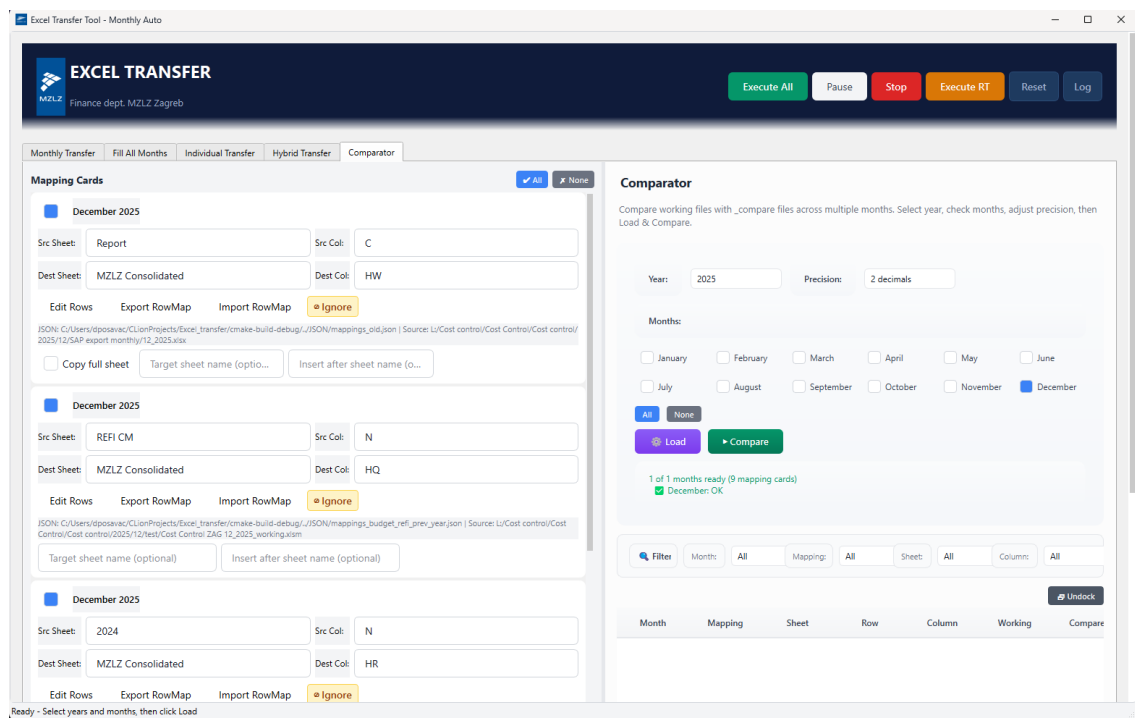


Figure 18: **Log panel** (toggled visible via the **Log** button in the header) showing detailed transfer execution logs. Each log entry includes a timestamp, severity level (INFO, WARNING, ERROR), and a descriptive message. The log captures all file loading operations, mapping executions, cell transfers, and any errors encountered. Users can scroll through the log history to troubleshoot issues or verify successful transfers. The log panel can be hidden at any time by clicking the **Log** button again.

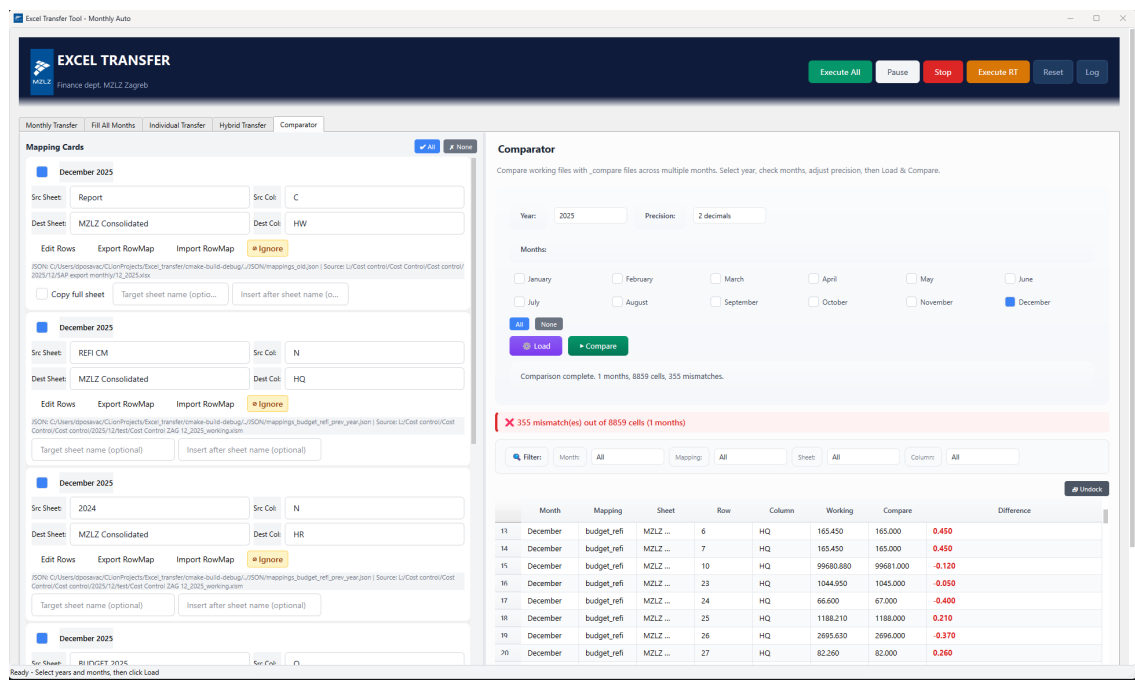


Figure 19: **Create Month File confirmation dialog** showing the source template file and destination paths for new month files. The dialog lists each month file that will be created (e.g., “May 2025”, “June 2025”, “July 2025”) along with its full destination path. The source template (e.g., “Cost Control ZAG 04_2025_working.xlsm”) is displayed at the top. Click **Yes** to proceed with file creation, or **No** to cancel. Already-existing files are automatically skipped to prevent accidental overwrites.

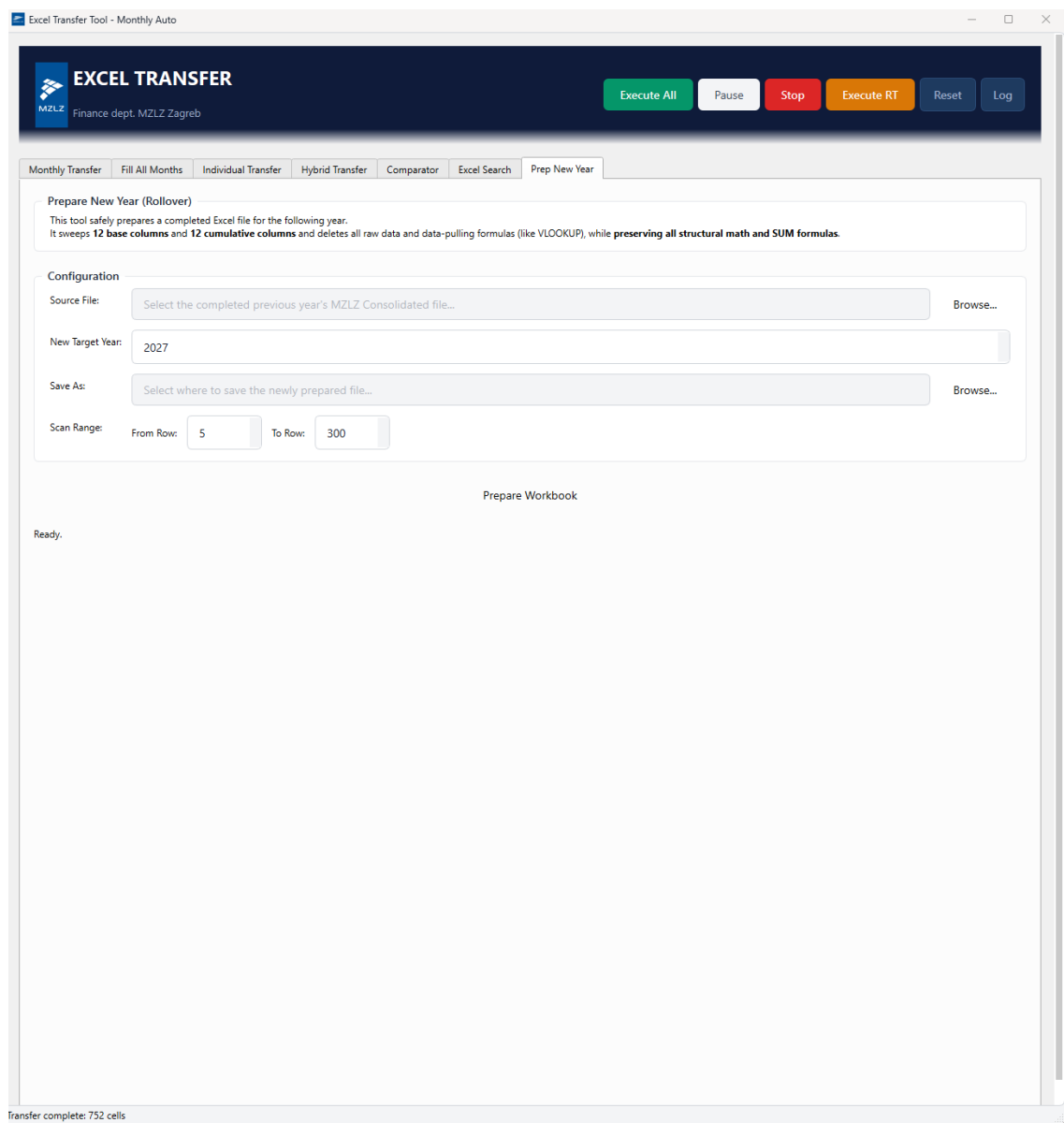


Figure 20: **File validation error dialog** displayed when the application detects a corrupt or invalid Excel file. The dialog shows the file path and error details (e.g., “not a zip file”, “missing workbook.xml”, “file is locked”). Common causes include: file open in Excel, corrupted ZIP structure, or incomplete download. The dialog offers options to **Replace** the file (browse for a valid replacement), **Skip** the file (continue without it), or **Cancel** the entire operation.

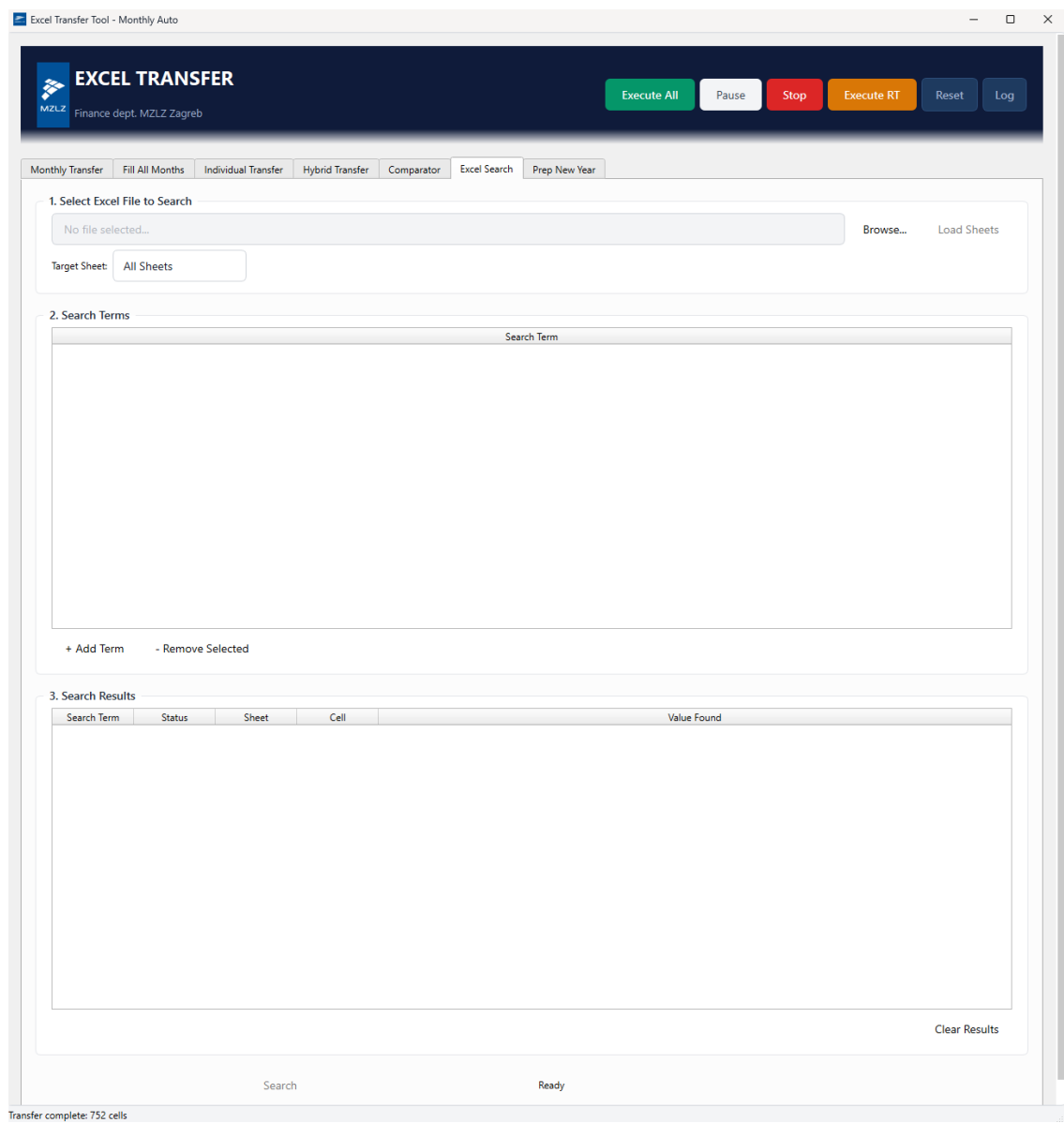


Figure 21: **Transfer completion summary** showing the final results after an Execute All or Fill All Months operation completes. The summary displays: total mappings processed, successful transfers, failed transfers, total cells transferred, and execution time. Any warnings or errors are listed below with clickable links to jump to the relevant log entries. The **Close** button dismisses the dialog. A green toast notification also appears at the bottom-center of the application window confirming successful completion.